2. The Physics and the Price

The electricity cost of orbital velocity is $0.53 per kilogram. Everything above that is hardware amortization, not physics.

Relation	What it means
$E = \frac{1}{2}v^2 = 32$ MJ/kg	At 8 km/s and $0.06/kWh: **$0.53 of grid electricity per kilogram**. Falcon 9 burns ~$200–300K of propellant for ~17–22.8 t to LEO, propellant alone costs ~20–25× KERAUNOS's total electricity spend per kilogram, on a vehicle that is 90% propellant by mass.
$a = v^2/2L$	Acceleration is set by velocity and length. 1,000 km at 8 km/s is 3.3 g, gentle enough for people. StarTram Gen-1 accepted 30 g on 130 km, cargo-only. On the Moon a 3 g orbital track is ~50 km. **Track length is the unlock.**
Kick stage: Isp 350 s, 20% mass	$\Delta v = 3{,}434 \times \ln(1/0.8) \approx$ **0.77 km/s** for circularization and trim at apogee. That is the only propellant aboard, ~80% of launched mass is payload, vs ~10% for a rocket.
Bends are forbidden at speed	Centripetal load is $a = v^2/R$, so a g-capped turn needs an enormous radius. KERAUNOS takes the rule to its conclusion: **every track is a single straight line, no terminal curve at all**. Straight means zero lateral g on the payload (Section 4 / Theoretical F).

```
Worked from the two relations above:
Energy floor: E = ½v² = ½ × (8,000 m/s)² = 32 MJ/kg = 8.9 kWh/kg ⇒ at $0.06/kWh = $0.53/kg to orbital speed
Track g-load: a = v²/2L, a 1,000 km track at 8 km/s ⇒ a = 8,000² / (2 × 10⁶) = 32 m/s² = 3.3 g (human-rated)
Why no bends: R = v²/a, turning at 4 km/s under an 8 g cap needs R = 4,000² / 78.5 = 204 km; so the track stays straight
```

Comparative launch costs (per kg to LEO)

System	Cost per kg	Economic bottleneck
Space Shuttle	$54,500 (full-program-cost basis)	Hand-built; massive refurbishment every flight
Falcon 9 (2026)	~$3,245 list (expendable basis); ~$630 est. internal	$74M list price (Feb 2026); reusable flights carry ~17–18 t, so real per-kg runs higher
Starship (projected)	$100–$1,000, **aspirational**	V3 flew once (Flight 12, May 22, 2026; payload deploy succeeded, booster mishap, fleet grounded pending FAA review). No reuse-economics data yet; chemical energy density is the hard floor
KERAUNOS	$20–$250 (target)	$0.53/kg electricity; the rest amortizes track capital, the honest unknown is tunnel+tube cost per km, which Phase 1 exists to measure
KERAUNOS + skyhook	toward $5	Kick propellant → ~2–5% when a rotating tether catches pods (Section 8)

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3. Why Now: The Signals, as of June 2026

The field validated itself publicly in the last 24 months. The window to enter at peer level is now.

- **SpaceX wants a lunar mass driver, said out loud, February 2026.** Following the company's lunar pivot (Feb 2, 2026), Musk described a mass driver "shooting AI satellites into deep space" at the Feb 11 all-hands, and reaffirmed it March 22 ("Mass drivers on the Moon!"). Honesty requires the caveat: these are founder statements, not a funded program of record, no contracts or milestones have been disclosed. But the largest launch company on Earth pointing at exactly this machine is both validation and a clock. A partner positioned ahead of that build doesn't compete, it supplies the architecture.
- **China is building EM launch hardware, targeting 2028.** Galactic Energy, CASIC, and the Ziyang municipal government are constructing an electromagnetic launch verification platform to throw a Ceres-2-class rocket to **above Mach 1** before ignition. Progress is real: a full-system ejection test of a 1.4 m rocket model (April 2025) and a matrix-switched segmented-drive trial at ~90% system efficiency (January 2026). It's launch-assist, the rocket still carries the full propellant penalty, but the supply chain and urgency are real.
- **The US military already funds this class of technology.** Auriga Space has raised $12.2M cumulatively (VC plus AFWERX, including a $1.25M Direct-to-Phase-II SBIR) for an EM launch-assist track, selling hypersonic test services as Phase-1 revenue, Prometheus (lab) → Thor (outdoor track, 2026) → Zeus (orbital). General Atomics EMALS (484 MJ flywheel storage, 45 s recharge) has launched aircraft from USS Gerald R. Ford since 2017 through a combat deployment. A 2023 AFOSR-filed report (Peterkin) explicitly recommends evolving EMALS for lunar launch, and a US SBIR program record describes a magnetically levitated sled driven to 7 km/s. The demand signal is public and funded.
- **The boom-shaping toolkit just went supersonic.** NASA's X-59 flew Oct 28, 2025 (67 min), expanded its envelope through spring 2026, and made its **first supersonic flight June 5, 2026** (Mach 1.1), with Mach 1.4 acoustic-validation points next, flying on a camera-vision cockpit the whole way. The quiet-shock shaping ASTRAPE borrows is being flight-proven right now.
- **The gap nobody fills:** every competitor is fighting the atmosphere, China's launch-assist, StarTram's paper tube, Auriga's track. Nobody is building the launcher where the atmosphere doesn't exist. KERAUNOS starts there; ASTERIA is the open lane, and Earth revenue (Chimborazo) funds the climb.

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4. The Chimborazo Project: BRONTE's Earth Demonstrator

Rev 2 replaces the sketch with a site study: real terrain, real geology, one straight line through the mountain.

Chimborazo, Ecuador. Summit 6,263 m, 1.47°S. The case still holds: the equator gives +465 m/s of Earth's rotation free (fire east); high exit altitude halves the air (at 5,327 m, density is ~54% of sea level, Rev 1 quoted 45%, which is the *pressure* ratio); equatorial launch reaches every inclination. What changed is the engineering, driven by three facts established in the Rev 2 site study:

- **Terrain (SRTM 30 m):** the mountain is a gentle 40-km shield with a concave western approach, no straight surface route to a summit-area exit exists. The answer is a **through-mountain line**: azimuth 095°, passing 0.5 km north of the summit.
- **Geology (IG-EPN, Bernard et al., Beauval et al.):** the andesite core is TBM-proven rock (Ecuador's own Coca Codo Sinclair bored 24.8 km with 9 m machines); the volcano is dormant-but-potentially-active (last eruption ~550 CE, ~1,000-yr recurrence); the governing hazard is seismic, the Pallatanga fault (1797 Riobamba, M~7.6) sets a ~0.4 g design basis, which drove A-frame piers, radial truss bracing, and re-trimmable track jacks. The route avoids the Riobamba debris-avalanche deposit entirely.
- **The straight-line rule, taken all the way:** at exit speed a track cannot bend without crushing the payload sideways, so BRONTE is a *single straight line at 6.3°* from portal to crest muzzle. We add no terminal curve to steepen the exit (Theoretical F shows why); straight means no lateral g, no centripetal "load wall," and no curve-failure case: the pod feels only its axial launch g. The price is track length, a steeper, straight grade is shorter for a given exit, which trims top speed. Both the geometry and that trade are derived below.

The line, in two builds, now a single straight grade

The muzzle is set at **5,327 m** on the upper eastern flank, **936 m below the 6,263 m summit**. This is the highest a straight 38 km line can reach on the real SRTM terrain: a straight line cannot meet the summit without a kilometre-tall viaduct over the lower flank, so the muzzle exits where the line surfaces on the upper cone and fires east over the drop.

$$\text{Straight track at fixed exit angle } \theta = 6.3° \ (\sin 6.3° = 0.1097)\text{: } L = \Delta h / \sin\theta$$
$$\text{Phase 1: } L = (5{,}327 - 3{,}791) / 0.1097 = 1{,}536 / 0.1097 = \mathbf{14{,}000\ m \approx 14.0\ km}$$
$$\text{Full line: } L = (5{,}327 - 1{,}109) / 0.1097 = 4{,}218 / 0.1097 = \mathbf{38{,}450\ m \approx 38\ km}$$
$$\text{Exit speed from } a = v^2/2L \Rightarrow v = \sqrt{2aL}\text{, cargo } a = 30\ g = 294\ m/s^2 \ (\text{local sound speed } a_s \approx 316\ m/s)\text{:}$$
$$\text{Phase 1: } v = \sqrt{2 \times 294 \times 14{,}000} = 2{,}869\ m/s = \mathbf{2.87\ km/s}\ (\text{Mach } 9.1),\ t = v/a = 9.8\ s$$
$$\text{Full line: } v = \sqrt{2 \times 294 \times 38{,}000} = 4{,}730\ m/s = \mathbf{4.73\ km/s}\ (\text{Mach } 15.0),\ t = 16.1\ s$$
$$\text{Energy/shot (15 t, full): } E = \tfrac{1}{2}mv^2 = \tfrac{1}{2} \times 15{,}000 \times 4{,}730^2 = 1.68 \times 10^{11}\ J = \mathbf{46.6\ MWh}$$

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	Phase 1, build first	Full line, the growth target
Length @ 6.3°	**14 km** (portal ~3,791 m, upper flank)	**~38 km** (west portal ~1,109 m, Chazojuán valleys)
Works	bored tunnel through the flank + crest bore + short truss muzzle	adds ~24 km of deep tunnel down the western slope
Exit	Same muzzle for both: **5,327 m**, **6.3° east, straight**, ~936 m below peak elevation	
Cargo (30 g)	2.87 km/s · **Mach ~9.1** · 9.8 s in tube	4.73 km/s · **Mach ~15.0** · 16.1 s
Human (3 g)	0.91 km/s · Mach ~2.9 · 31 s	1.49 km/s · Mach ~4.7 · 51 s
Per shot (15 t gross)	17.1 MWh · 12.7 GW peak	46.6 MWh · 20.8 GW peak, trackside storage; grid sees ~30 MW × ~52 min recharge

Could the line go faster? The velocity ladder over the 38 km line

The full line runs cargo at 30 g, which over 38 km is 4.73 km/s (Mach 15.0), one rung on this ladder. Exit speed is set by length and g-load ($v = \sqrt{2aL}$); here is what dialing the same 38 km line to 4, 6, and 8 km/s would take, and whether the g-load makes it possible.

```
Over L = 38 km: a = v²/2L, g = a/9.81, t = 2L/v, F = ma (15 t), E = ½mv²
4 km/s: a = 4,000²/(2·38,000) = 211 m/s² = 21.5 g · t = 19 s · F = 3.2 MN · E = 33 MWh
6 km/s: a = 6,000²/76,000 = 474 m/s² = 48.3 g · t = 12.7 s · F = 7.1 MN · E = 75 MWh
8 km/s: a = 8,000²/76,000 = 842 m/s² = 85.8 g · t = 9.5 s · F = 12.6 MN · E = 133 MWh
```

Target exit (over 38 km)	g-load	Time in tube	Force (15 t)	Energy/shot (15 t · 20 t)	Possible for
4 km/s	21.5 g	19 s	3.2 MN	33 · 44 MWh	most cargo & satellites
6 km/s	48.3 g	12.7 s	7.1 MN	75 · 100 MWh	g-hardened / bulk only
8 km/s ≈ orbital	85.8 g	9.5 s	12.6 MN	133 · 178 MWh	inert bulk only (fuel, water, metal)

Verdict: **all three are physically possible**, even 8 km/s, essentially orbital velocity with no kick stage, at 86 g (well inside the ~15,000 g gun-hardened electronics survive). The limit is the payload, not the track: 4 km/s (20 g) suits most cargo, 6 km/s needs hardening, 8 km/s only inert bulk. Because $g \propto 1/L$, a longer track buys the same speed at lower g, the 1,000 km-class injector (§9) reaches 8 km/s at just 3.3 g, the human-rated regime.

The tube itself: Ø4.0 m flight bore inside a Ø6.0 m ring-stiffened vacuum vessel inside a Ø12 m TBM bore, driven by a **360° stator ring** (24 REBCO sectors per ring, hoops every 0.6 m). On a straight track the ring carries no curve side-load; it stays full-circumference for two reasons: **distributed full-hull thrust** (the reaction sleeve is driven along its whole length, so the pod never accumulates the axial compression that would buckle a tail-pushed projectile) and **precise magnetic centering** of a Mach-15 vehicle inside a 200 mm vacuum gap. The payload bay is **Ø3.2 m × 8 m** (Cygnus-module class) inside the 23 m ASTRAPE fairing-sled. Vacuum isolation gates every 500 m; 20 K/80 K cryo headers full-length with cooler stations every 500 m.

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The honest open issues, on the record: El Arenal piers reach ~590 m (Millau's record is 343 m); the plasma window is proven only at centimeter scale and the Ø4 m bore costs 2.8× the baseline window power, it gets the first R&D dollar; the whole massif sits inside the Reserva de Producción de Fauna Chimborazo under Ecuador's rights-of-nature constitution, the legal path may be the hardest engineering on the sheet; and Mach-15 exits are loud regardless of shaping, altitude and the empty eastern corridor are the real mitigations, and test recovery overflies the inter-Andean valley like any sounding-rocket range.

Revenue from day one

DoD hypersonic test services: sounding rockets are single-shot, wind tunnels don't hold real Mach 6+ conditions for minutes, and no facility on Earth fires Mach-7-class test articles from 5,327 m on the equator several times a day. At $50K–$500K per run, one run per week covers operations. The demonstrator is not a cost center, it is a product, and every shot also retires risk on the plasma window, the aero-shell, and the cost-per-kilometer number the full injector's economics live or die on.

The sonic-boom problem, possibly the real showstopper

A Mach-15 body leaving the muzzle at 5,327 m is loud in a way altitude and shaping cannot fully fix. This may be the constraint that decides the site, or kills a populated one.

```
Reference (Concorde): ~1.94 psf (~93 Pa) overpressure, ~105 PNLdB, at Mach 2 and ~16 km altitude.
BRONTE exits at Mach 15 and only 5,327 m (~5.3 km), ~3.0× closer to the ground and ~7.5× faster.
Boom overpressure rises as the exit gets lower and faster (∝ ~altitude^-3/4, weakly with Mach):
  altitude factor (16/5.3)^3/4 ≈ 2.3× · Mach factor ((15²−1)/(2²−1))^1/8 ≈ 1.7× ⇒ ~4× Concorde
⇒ order 8–14 psf (~400–700 Pa) near the track, window-rattling, not merely audible, repeated
several times a day.
```

Honest verdict. For a populated equatorial site this is a first-order public-acceptance risk, not a footnote: Riobamba (~170,000 people, ~30 km SE) and Ambato sit within the acoustic footprint, and a launcher firing several times a day would subject them to repeated hypersonic booms. That compounds the rights-of-nature reserve problem, together they may make a Chimborazo build politically impossible regardless of the physics. Mitigations help but do not eliminate it: fire *east* over the empty Amazon-facing corridor, exploit the altitude, shape the exit body, cap the firing cadence and hours, and, the real lever, be willing to relocate the muzzle/corridor away from inhabited valleys. The boom, more than the rock, may choose where (or whether) this gets built. (Concorde figures: NASA sonic-boom data; overpressure scaling is first-order estimate.)

What it costs, and what it could earn

Order-of-magnitude only, the whole point of Phase 1 is to *measure* the cost-per-kilometer nobody has.

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CAPEX, Phase 1 (14 km demonstrator):
Tunnel: ~14 km of Ø12 m bore at ~$200–400M/km (large-bore TBM; cf. Gotthard $10.2B / ~$90M/km over
57 km of mostly smaller works; modern large urban TBM ≈ €500M/km) → ~$3–6B
Launcher: 360° REBCO ring (~23,000 hoop-rings on 14 km), 20 K cryoplant, vacuum vessel, plasma
window, pulsed-power storage for a ~13 GW peak, a fusion-magnet-class program → ~$6–14B
Power plant + storage farm + site/seismic works → ~$1–3B
Phase-1 total ≈ $10–20B (cf. Gotthard tunnel $10.2B · ITER >$22B · one SLS launch $4.1B · Artemis
~$100B to date)
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> **REVENUE:**
> Phase 1 (Mach 9, *sub-orbital* test platform): DoD hypersonic test services \$50K–\$500K/run × ~1–5 runs/day → **~\$70–350M/yr**, covers operations, **does not repay the build alone**. Phase 1 is a risk-retirement bet.
> Full line / injector (orbital): launch cost ~\$20–250/kg vs market ~\$3,000/kg. At the full injector's 29,000 t/yr, even selling at **\$200/kg ⇒ ~\$5.8B/yr** gross, a fraction of that repays the build in years *if* orbital demand and the per-km cost hold.

The honest bottom line: Phase 1 is a **\$10–20B bet** whose return is the *data* and the orbital business it unlocks, not test fees. It pencils out only if (a) the per-km tunnel+tube cost lands near the low end, (b) orbital launch demand is real, and (c) the sonic-boom and legal constraints permit operation at all. (Tunnel costs: Gotthard Base Tunnel financials; large-bore TBM unit costs. Market launch price: Falcon 9 2026 ~\$3,245/kg.)

The BRONTE Chimborazo Design Map (Annex B) follows this page, grouped here with the Chimborazo project: route map on real terrain, both build profiles at true 1:1 scale, tunnel & tube sections, the 360° drive ring, ASTRAPE in three views, the cable runs, and the engineering charts. Every figure on it re-derived and audited June 2026. (The SELENE South Pole Design Map, Annex A, is grouped earlier with Section 1.) Each large panel is laid out on its own page and rotated to read at full size.

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